Turning-tool stories

How Drill Bits Work

Name _____ Date _____

Show your thinking with numbers, marks, or words.

1. Little counts 24 screws in a jar. Big adds 17 more. How many screws are in the jar now?

24 + 17 = _____

Show your thinking.

2. Big has 63 screws and puts 28 into a box. How many screws stay in the jar?

63 − 28 = _____

Show your thinking.

3. Little counts 16 turns of a hand crank, then 27 more turns. How many turns does Little count altogether?

16 + 27 = _____

Show your thinking.

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4. There are 54 pencil curls on a tray. Little brushes 22 into a paper bag. How many curls remain on the tray?

54 − 22 = _____

Show your thinking.

5. Big turns an egg-beater handle 38 times, then 15 more times. How many handle turns does Big make in all?

38 + 15 = _____

Show your thinking.

6. A sand tray holds 71 small scoops of sand. A hand auger lifts out 34 scoops. How many scoops of sand remain?

71 − 34 = _____

Show your thinking.

WONDER DAILY

How Drill Bits Work

Follow the spiral

Little found a spiral on a drill bit. As the bit turns, the cutting lips at its tip shave off tiny chips.

A thread in soap

Big turned a screw into soap. Its thread wound around like a tiny ramp, helping the screw move in with each turn.

A way out

Chips need a way out of the hole. The drill bit's spiral grooves, called flutes, carry them up as it turns.

Around and up

An auger has a broad spiral. As it turns in the soil, it lifts loose dirt up and out of the hole.

Water takes a ride

Big turned the Archimedes screw. Pockets of water rode up the sloping tube and spilled into a cup higher than the bowl.

Keep a question

Little drew a spiral beside the turning tools. "Why does a spiral move things when it spins?" The notebook had room to wonder.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

Give the soap test its words

How Drill Bits Work · Curiosity day

Name _____ Date _____

Say one clear command, then draw or write it.



A spiral to follow



A thread in the soap



A way out for chips

Draw the screw in the soap. Add a curl beside the thread.

turn · slowly · screw · soap · thread · curl · spiral · notice

First, … tell how to set the soap on a tray

Then, … tell how to turn the screw slowly

Look for, … tell where to look for the curls

Does your command tell what to do, using a word such as Turn, Look, or Notice?

How Drill Bits Work

01 Which part makes the hole?



The cutting lips at the tip shave off chips.

Look closer. Look at the cutting edges at the tip.

A little explanation. A twist drill bit is a metal rod with spiral grooves. As it turns and moves into the material, the cutting lips at its tip shave off tiny bits called chips.

Try this. Trace the tip in the picture, then follow one spiral groove.

02 Where do the chips go?



Flutes carry chips up and out of the hole.

Look closer. Follow the chips from the bottom of the hole to the top.

A little explanation. The spiral grooves are called flutes. They help carry chips up and out as the bit turns, keeping the hole clear. The cutting lips make chips; the flutes give them a way out.

Try this. Use the diorama to follow one chip up the flute.

03 What can we notice by turning slowly?



A screw thread helps a turning screw move into the soap.

Look closer. Look for a curl beside the screw thread.

A little explanation. A screw has a spiral ridge called a thread. It acts like a ramp wrapped around the rod. Turning against the material helps the screw move in. A screw thread is a raised ridge; a drill flute is a groove.

Try this. With Big close by, hand-turn a screw slowly into soap or clay on a tray and look at the curls. Power drills are tools for adults; kids do not use them. Big stays close. Try turning a pencil in an enclosed hand sharpener, too.

04 What else can a turning spiral move?



An Archimedes screw lifts water to a higher place.

Look closer. Water rides from a low bowl to a higher cup.

A little explanation. An auger lifts loose soil or ice from a hole. An Archimedes screw lifts pockets of water along a sloping tube. A spiral slopes along its length, so turning against material can push or carry that material along. It needs something to work against, not just empty space.

Try this. Wrap a ribbon around a pencil and follow its sloping path with your eyes.